

CHAPTER 8

Additional Vocations



0916CH08

Sheet Metal Work

Sheet metal work involves shaping thin metal sheets (such as steel, aluminum and copper) into useful products by processes including cutting, bending, forming, joining and finishing. This work requires careful measurement, correct tool selection, safe handling of sharp edges, and attention to quality and finish.



Key words

Forming:
shaping
metal without
cutting it

Burr: sharp
unwanted
edge after
cutting

Rivet:
fastener
used to join
metal sheets

Gauge:
thickness
of a sheet

Item/Material name	Description/Use
GI/Aluminium sheet	Main material for fabrication
Fasteners/Rivets	Joining sheet metal parts
Cardboard/Paper	Prototype
Tin snips/Shears	Manual cutting of sheet metal
Hammer/Mallet	Shaping and forming
Files/Grinder	Smoothing sharp edges
Drill/Hand tools	Making holes
Measuring scale/Tape	Accurate measurement
Marking pen/Scriber	Marking layout
Safety gear	Gloves, goggles, apron

Quality parameters

Accurate dimensions: proper measurement and marking

Smooth edges: burrs removed and edges finished

Strong joints: secure riveting or joining

Proper shape: correct bends and alignment

Minimum waste: efficient use of material and reuse of scrap

Safety parameters

Wear safety gloves, goggles, ear plugs and apron while working

Use cutting and forming tools only as demonstrated

Keep fingers away from cutting and bending zones

Maintain a clean and organised workspace

Dispose waste as per protocol

Key steps in the process

1. **Scoping:** Choose a simple item, such as a dustpan, dustbin, chalk box and box clamps. Decide the quantity to be made.
2. **Design and draw:** Draw simple sketches of the product. Mark dimensions and decide the forming technique.
3. **Make a prototype:** Create a prototype using cardboard to get an idea of the design, especially folds and joints.
4. **Select materials and tools:** Select suitable sheet metal and cutting tools (tin shears).
5. **Prepare the product:** Mark on sheet using scribe; cut the metal; use punching tools, riveting or folding and hammering sheet metal, or soldering as required.
6. **Finish the product:** Smooth surfaces, trim edges and apply colours.

Plumbing

Plumbing is the work of installing, maintaining and repairing systems that carry water, liquids, gases, fluids or waste through pipes. Plumbing includes planning pipe routes, selecting suitable materials, cutting and joining pipes, fitting taps and valves, checking water flow and repairing leakages.



Keywords

Valve:
device used to
control water
flow

Joint:
connection point
between two
pipes or fittings

Flow:
movement of
water through a
pipe system

Item/Material name	Description/Use
Pipes (PVC/GI/CPVC)	Carrying water
Pipe fittings	Couplings, elbows, tees, end caps
Taps/Valves	Controlling water flow
Pipe wrench/Spanner	Tightening and loosening joints
Hacksaw/Pipe cutter	Cutting pipes to length
Thread seal tape/Adhesive	Preventing leakage at joints
Plunger	Clearing blocked drains
Measuring tape/Spirit level	Measuring and alignment
Cloth/Bucket	Cleaning and water collection
Safety gloves/Goggles	Personal safety

Quality parameters

Leak-free joints: No dripping or seepage from pipes or fittings

Proper alignment: Pipes are laid straight and level

Secure fittings: Valves and fixtures are tightly fixed

Smooth water flow: No blockages or air locks

Minimum waste: Efficient use of pipes, fittings and water

Safety parameters

Wear basic protective equipment, such as gloves and safety glasses

Keep hands dry while handling tools and fittings

Ensure the work area is clean and free of obstacles

Dispose waste as per protocol

Key steps in the process

1. **Scoping:** Observe plumbing systems through site visits, videos or interaction with plumbers working in homes, schools, farms or in the community. Choose a simple and safe task, such as fixing a leaking tap, replacing a joint, clearing a blocked drain, or installing a short pipe section.
2. **Design and draw:** Study the water flow path and prepare a simple sketch showing pipes, joints and fixtures.
3. **Prepare detailed plan:** Prepare a step-by-step plan covering material selection, tool use, safety checks and execution.
4. **Select materials and tools:** Identify suitable pipes, fittings and tools based on the task and availability.
5. **Carry out the work:** Ensure water supply is shut off. Measure, cut and join pipes; tighten fittings; apply sealing material; and fix fixtures carefully.
6. **Finish the work:** Restore water supply and check for leaks, blockages or improper flow.



Food Processing

Food Processing involves transforming raw food materials into safe, usable and longer-lasting food products through simple activities, such as cleaning, cutting, drying, cooking, fermenting and preserving.



Keywords

Raw material: unprocessed food such as grains, fruits, vegetables or milk

Processing: clean, cut, cook, preserve raw material

Preservation: methods used to increase shelf life of food

Item/Material name	Description/Use
Raw food materials	Grains, fruits, vegetables, pulses, etc.
Clean water	Washing and processing food
Knives/Peelers	Cutting and cleaning ingredients
Chopping board	Safe food preparation
Bowls/Containers	Mixing and holding ingredients
Stove/Heating source	Cooking or roasting (teacher-supervised)
Drying trays/Cloth	Drying food materials
Weighing scale/Cups	Measuring quantities
Packaging material	Jars, pouches, paper labels
Cleaning supplies	Cloth, soap, waste bins

Quality parameters

Freshness: Raw materials are clean and suitable for use

Hygiene: Processing is done in a clean and orderly manner

Safety parameters

Wash hands thoroughly before handling food

Keep work surfaces, tools and containers clean

Consistency: Product has uniform texture, taste and appearance

Safety: Food is properly cooked, dried or preserved

Minimum waste: Efficient use of food, water and materials

Use knives, peelers and heat sources carefully

Avoid cross-contamination between raw and processed food

Store ingredients and finished products properly

Dispose waste as per protocol

Key steps in the process

1. **Scoping:** Choose a simple product, such as roasted grains, dried vegetables, pickle, papad or snack item.
2. **Follow a recipe:** List ingredients required, and check their quality and freshness; organise work area for cleaning, preparation, processing and packaging.
3. **Prepare ingredients:** Wash, peel, cut or sort food materials following hygiene practices.
4. **Carry out the work:** Perform selected processing techniques such as drying, roasting, cooking, or fermenting using safe methods; note changes in colour, texture, smell, and taste during processing, and take necessary actions.
5. **Finish the work:** Check the product for cleanliness, safety and consistency; pack the food safely using clean containers or pouches. Add simple labels, if required.

Furniture Making

Furniture making involves designing and creating useful objects, such as stools, tables, shelves, benches, storage units and simple seating. Furniture making combines practical skills with creativity, while promoting responsible use of materials, reuse of waste wood or metal, and careful workmanship.



Keywords

Joinery: methods used to join two or more parts of furniture

Seasoned wood: wood that has been dried to reduce moisture

Finishing: final surface treatment such as sanding, polishing or painting



Item/Material name	Description/Use
Wood/Bamboo/Engineered boards	Main material for furniture
Metal rods/Pipes (optional)	Frames or supports
Measuring tape/Steel rule	Measuring dimensions
Try square/Spirit level	Checking right angles and alignment
Hand saw/Chisel/Plane	Cutting and shaping material
Drill/Screwdriver	Making holes and fixing joints
Bench vice/Clamps	Holding parts securely
Hammer/Mallet	Assembling joints
Fasteners (nails, screws, and bolts)	Joining parts
Sandpaper/Paint/Varnish	Finishing and protection
Waste bins	Safe disposal of scrap

Quality parameters

Strength and stability: Furniture can safely support its intended load

Accuracy: Measurements and joints fit properly

Finish: Surfaces are smooth and edges are safe

Useability: Furniture meets the user's need comfortably

Material efficiency: Recycling, minimum wastage and reuse of leftovers

Safety parameters

Wear basic protective equipment, such as gloves and safety glasses when required

Keep tools sharp, clean and stored properly after use

Keep the workspace clean and free of obstacles

Dispose waste as per protocol

Key steps in the process

1. **Scoping:** Choose a simple product, such as a stool small table, shelf or box.
2. **Design and draw:** Draw a simple sketch with dimensions; decide materials, joints and basic layout; digital tools may be used if available.
3. **Prepare a prototype:** Make a test model using cardboard or paper to check shape, joints, stability, etc.
4. **Select materials and tools:** Choose locally-available materials and suitable tools, considering strength, cost and safety.
5. **Prepare the product:** Measure, mark, cut and shape parts carefully; join parts using nails, screws, glue or joints. Check alignment and stability.
6. **Finish the product:** Sand surfaces and apply paint, polish or varnish for protection and appearance.

Pottery

Pottery involves preparing clay, shaping it using simple techniques, drying it carefully, strengthening it through firing or hardening, and finishing the surface for use or decoration.



Keywords

Wedging: kneading clay to remove air bubbles and ensure even moisture

Firing: heating clay in a kiln or open fire to make it hard and durable

Slip: clay mixed with water, used for joining parts or surface decoration

Item/Material name	Description/Use
Clay (local)/Terracotta	Main material for pottery
Water	Softening clay and making slip
Sand/Grog	Improving strength and reducing cracks
Rolling pin/Paddles	Shaping clay
Wire cutter	Cutting clay lumps
Carving tools	Trimming and decoration
Sponges/Brushes	Smoothing and finishing
Moulds	Creating uniform shapes
Drying boards/Trays	Safe drying of products
Apron/Gloves	Personal safety and cleanliness

Quality parameters

Material quality: clay is clean, well-prepared and suitable for the product

Uniform thickness: product walls are even and balanced

Safety parameters

Use tools such as wire cutters and carving tools carefully

Keep hands, feet and work surfaces clean to avoid slipping



Stability: base is flat and the product stands firmly

Finishing: surface is smooth and free from cracks

Minimum waste: clay and water are reused wherever possible

Follow safety instructions during firing or hardening processes

Avoid inhaling clay dust; clean surfaces with wet cloths

Wear basic protective gear, such as apron or gloves if required

Dispose waste as per protocol

Key steps in the process

1. **Scoping:** Choose a simple item, such as a *diya*, bowl, cup, tile or planter. Decide the quantity to be made.
2. **Design and draw:** Draw simple sketches of the product. Mark dimensions and decide the forming technique.
3. **Make a prototype:** Create a small test piece to observe cracking, drying behaviour and strength.
4. **Select materials and tools:** Select suitable clay; knead and wedge it to remove air and control moisture. Procure tools required, based on the process you have selected.
5. **Prepare the product:** Use pinch, coil, slab or mould techniques. Join parts using slip and maintain even thickness; allow slow and shaded drying. Monitor for cracks or warping; harden the product through kiln firing, open firing or extended sun drying, following the safety norms.
6. **Finish the product:** Smooth surfaces, trim edges and apply simple textures or natural colours.
7. **Strengthen the product:** Harden the product through kiln firing, open firing or extended sun drying, following safety norms.
8. **Finish and decorate:** Smooth surfaces, trim edges and apply simple textures or natural colours.

